

1. GitHub link:

<https://github.com/hnguy130/CSE-464-2025-hnguy130>

2. Compiling Java project using mvn package

```

CSE464_GraphManager -- zsh -- 157x55
hodangkhoanguyen@MacBook-Pro-cua-Ho-8 CSE464_GraphManager % mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.asu:CSE464_GraphManager >-----
[INFO] Building CSE 464 Course Project 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ CSE464_GraphManager ---
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ CSE464_GraphManager ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ CSE464_GraphManager ---
[INFO] Copying 0 resource from src/test/resources to target/test-classes
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ CSE464_GraphManager ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE464_GraphManager ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running GrapherTest
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
-----
remove nodes test
-----
Graph file to be parsed with: test.txt
Node a has been removed from graph
Node b has been removed from graph
Graph information written to: output2.txt
-----
remove edge test
-----
Graph file to be parsed with: test.txt
Edge from e to f has been removed from graph
Graph information written to: output3.txt
-----

CSE464_GraphManager -- zsh -- 157x61
-----
remove node test
-----
Graph file to be parsed with: test.txt
Node a has been removed from graph
Graph information written to: output1.txt
-----
search_wrong test
-----
Graph file to be parsed with: test.txt
-----
search_correct test
-----
Graph file to be parsed with: test.txt
BFS: a->e->g
DFS: a->e->f->g
-----
remove wrong edge test
-----
Graph file to be parsed with: test.txt
-----
remove wrong node test
-----
Graph file to be parsed with: test.txt
Node a has been removed from graph
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.073 s -- in GrapherTest
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ CSE464_GraphManager ---
[INFO] Building jar: /Users/hodangkhoanguyen/Documents/ASU/junior/CSE 464/CSE-464-2025-hnguy130/CSE464_GraphManager/target/CSE464_GraphManager-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.899 s
[INFO] Finished at: 2025-03-26T03:16:07-07:00
[INFO]
hodangkhoanguyen@MacBook-Pro-cua-Ho-8 CSE464_GraphManager %
```

3. Features

a) removeNode(String node)

```
Grapher.java
478 }
479 }
480
481 nodeList.sort((a, b) -> a.name().value().compareTo(b.name().value()));
482 edgeList.sort((a, b) -> a.from().name().value().compareTo(b.from().name().value()));
483 }
484
485 public static void main(String[] args) throws IOException {
486
487     Grapher grapher = new Grapher();
488     String test = "test.txt";
489     grapher.parseGraph(test);
490     grapher.outputGraph("graphData.txt");
491
492     grapher.removeNode("b");
493
494     // String[] nodes = {"a", "b"};
495     // grapher.removeNodes(nodes);
496
497     // grapher.removeEdge("a", "e");
498
499     // grapher.GraphSearch("a", "g", Algorithm.BFS);
500     // grapher.GraphSearch("a", "g", Algorithm.DFS);
501
502     grapher.outputGraph("graphOutput.txt");
503
504 }
505 }
```

```
graphData.txt
1 Graph data:
2
3 Node amount: 8
4 Label of nodes: a , b , c , d , e , f , g , h
5
6 Edge amount: 10
7 Label of edges:
8 a -> b
9 a -> e
10 b -> c
11 c -> d
12 d -> a
13 e -> f
14 e -> g
15 f -> g
16 f -> h
17 g -> h
18
```

```
graphOutput.txt
1 Graph data:
2
3 Node amount: 7
4 Label of nodes: a , c , d , e , f , g , h
5
6 Edge amount: 8
7 Label of edges:
8 a -> e
9 c -> d
10 d -> a
11 e -> f
12 e -> g
13 f -> g
14 f -> h
15 g -> h
16
```

```
Console
<terminated> Grapher [Java Application] /Library/Java/JavaVirtualMachines/temurin-21.jdk/Contents/Resources/Java/JavaRuntimeLibraries/rt.jar
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See https://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Graph file to be parsed with: test.txt
Graph information written to: graphData.txt
Node b has been removed from graph
Graph information written to: graphOutput.txt
```

b) removeNodes(String[] nodes)

The screenshot shows the Eclipse IDE with the following components:

- Grapher.java** (lines 478-505):

```
478 }  
479 }  
480 }  
481 nodeList.sort((a, b) -> a.name().value().compareTo(b.name().value()));  
482 edgeList.sort((a, b) -> a.from().name().value().compareTo(b.from().name().value()));  
483 }  
484 }  
485 public static void main(String[] args) throws IOException {  
486 }  
487 Grapher grapher = new Grapher();  
488 String test = "test.txt";  
489 grapher.parseGraph(test);  
490 grapher.outputGraph("graphData.txt");  
491 }  
492 // grapher.removeNode("b");  
493 }  
494 String[] nodes = {"a","b"};  
495 grapher.removeNodes(nodes);  
496 }  
497 // grapher.removeEdge("a","e");  
498 }  
499 // grapher.GraphSearch("a", "g", Algorithm.BFS);  
500 // grapher.GraphSearch("a", "g", Algorithm.DFS);  
501 }  
502 grapher.outputGraph("graphOutput.txt");  
503 }  
504 }  
505 }
```
- Console**:

```
<terminated> Grapher [Java Application] /Library/Java/JavaVirtualMachines/tomurin-21.jdk/Contents/Resources/Java/JavaRuntimeLibraries/rt.jar:1: SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".  
SLF4J: Defaulting to no-operation (NOP) logger implementation  
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.  
Graph file to be parsed with: test.txt  
Graph information written to: graphData.txt  
Node a has been removed from graph  
Node b has been removed from graph  
Graph information written to: graphOutput.txt
```
- graphData.txt**:

```
1 Graph data:  
2  
3 Node amount: 8  
4 Label of nodes: a , b , c , d , e , f , g , h  
5  
6 Edge amount: 10  
7 Label of edges:  
8 a -> b  
9 a -> e  
10 b -> c  
11 c -> d  
12 d -> a  
13 e -> f  
14 e -> g  
15 f -> g  
16 f -> h  
17 g -> h  
18
```
- graphOutput.txt**:

```
1 Graph data:  
2  
3 Node amount: 6  
4 Label of nodes: c , d , e , f , g , h  
5  
6 Edge amount: 6  
7 Label of edges:  
8 c -> d  
9 e -> f  
10 e -> g  
11 f -> g  
12 f -> h  
13 g -> h  
14
```

c) removeEdge(String from, String to)

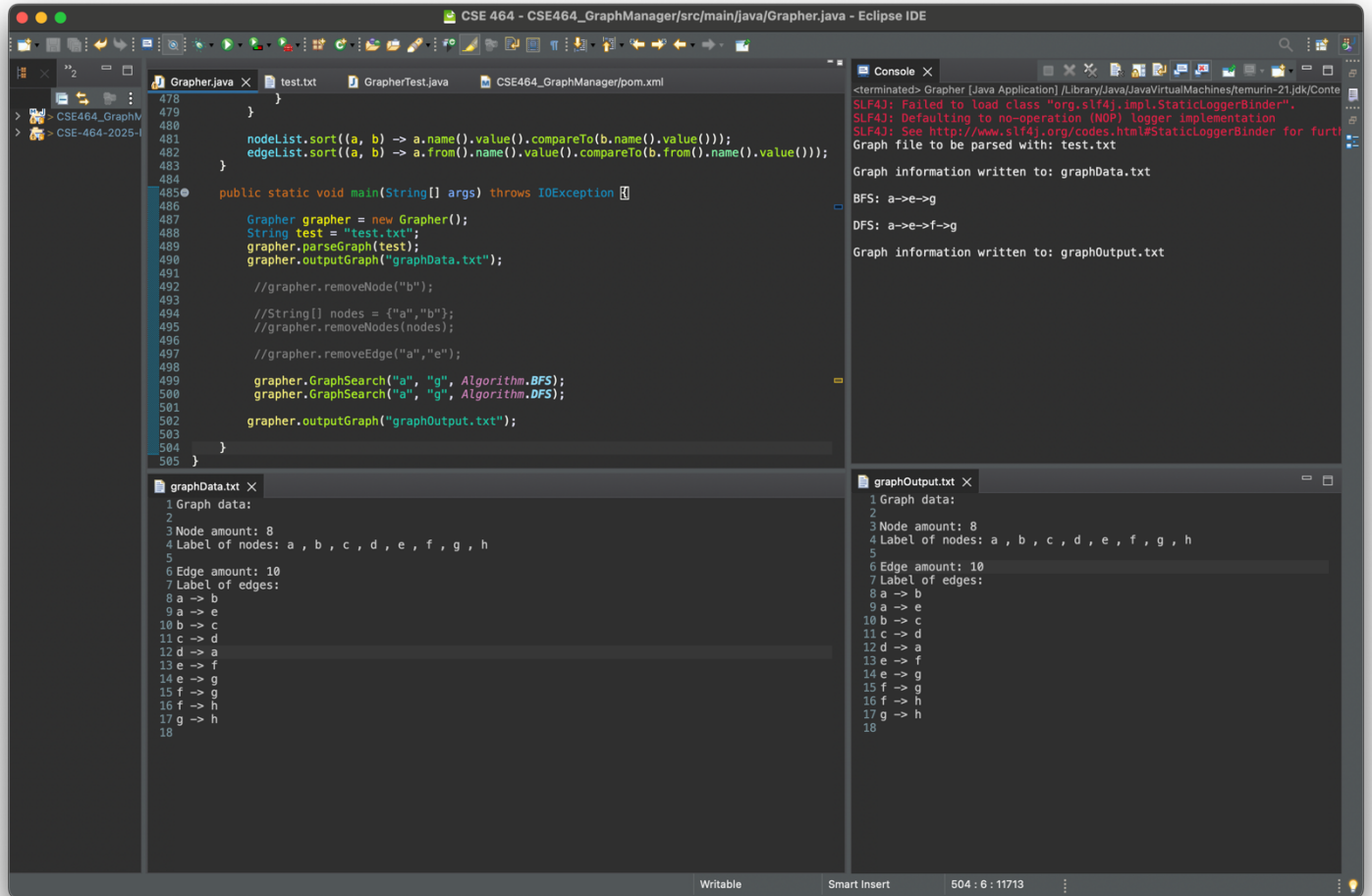
The screenshot shows the Eclipse IDE with the following components:

- Grapher.java:** Contains the `removeEdge` method and a `main` method. The `main` method creates a `Grapher` object, parses a graph from `test.txt`, outputs it to `graphData.txt`, removes node `"b"`, removes edge `"a" -> "e"`, and outputs the updated graph to `graphOutput.txt`.
- graphData.txt:** Contains the initial graph data:

```
1 Graph data:
2
3 Node amount: 8
4 Label of nodes: a , b , c , d , e , f , g , h
5
6 Edge amount: 10
7 Label of edges:
8 a -> b
9 a -> e
10 b -> c
11 c -> d
12 d -> a
13 e -> f
14 e -> g
15 f -> g
16 f -> h
17 g -> h
18
```
- graphOutput.txt:** Contains the graph data after removing node `"b"` and edge `"a" -> "e"`:

```
1 Graph data:
2
3 Node amount: 8
4 Label of nodes: a , b , c , d , e , f , g , h
5
6 Edge amount: 9
7 Label of edges:
8 a -> b
9 b -> c
10 c -> d
11 d -> a
12 e -> f
13 e -> g
14 f -> g
15 f -> h
16 g -> h
17
```
- Console:** Shows the output of the application, including the message "Edge from a to e has been removed from graph" and "Graph information written to: graphOutput.txt".

d) graphSearch(String from, String to, Algorithm algo(Algorithm.DFS or Algorithm.BFS))



```
Grapher.java
478 }
479 }
480
481 nodeList.sort((a, b) -> a.name().value().compareTo(b.name().value()));
482 edgeList.sort((a, b) -> a.from().name().value().compareTo(b.from().name().value()));
483 }
484
485 public static void main(String[] args) throws IOException {
486
487     Grapher grapher = new Grapher();
488     String test = "test.txt";
489     grapher.parseGraph(test);
490     grapher.outputGraph("graphData.txt");
491
492     //grapher.removeNode("b");
493
494     //String[] nodes = {"a","b"};
495     //grapher.removeNodes(nodes);
496
497     //grapher.removeEdge("a","e");
498
499     grapher.GraphSearch("a", "g", Algorithm.BFS);
500     grapher.GraphSearch("a", "g", Algorithm.DFS);
501
502     grapher.outputGraph("graphOutput.txt");
503 }
504 }
505 }
```

```
graphData.txt
1 Graph data:
2
3 Node amount: 8
4 Label of nodes: a, b, c, d, e, f, g, h
5
6 Edge amount: 10
7 Label of edges:
8 a -> b
9 a -> e
10 b -> c
11 c -> d
12 d -> a
13 e -> f
14 e -> g
15 f -> g
16 f -> h
17 g -> h
18
```

```
graphOutput.txt
1 Graph data:
2
3 Node amount: 8
4 Label of nodes: a, b, c, d, e, f, g, h
5
6 Edge amount: 10
7 Label of edges:
8 a -> b
9 a -> e
10 b -> c
11 c -> d
12 d -> a
13 e -> f
14 e -> g
15 f -> g
16 f -> h
17 g -> h
18
```

```
Console
<terminated> Grapher [Java Application] /Library/Java/JavaVirtualMachines/temurin-21.jdk/Contents/Resources/Java/JavaRuntimeLibraries/rt.jar
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Graph file to be parsed with: test.txt
Graph information written to: graphData.txt
BFS: a->e->g
DFS: a->e->f->g
Graph information written to: graphOutput.txt
```

4. Links:

- branch main: <https://github.com/hnguy130/CSE-464-2025-hnguy130>
- branch bfs : <https://github.com/hnguy130/CSE-464-2025-hnguy130/tree/bfs>
- branch dfs : <https://github.com/hnguy130/CSE-464-2025-hnguy130/tree/dfs>

- merge bfs : <https://github.com/hnguy130/CSE-464-2025-hnguy130/commit/9da43690281b49a9207270eb479a1985c15e3cc4>
- merge dfs : <https://github.com/hnguy130/CSE-464-2025-hnguy130/commit/e938717db59148a4972606aa782567a716d5c521>